

TOPIC 2: LINEAR SYSTEMS

1. INTRODUCTION

A *system of m linear equations with n unknowns* is given by

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{cases}$$

where a_{ij} and b_k are given real numbers and $x_1, \dots, x_n$ are the unknowns or variables of the system.

A linear system can be written in matrix form as $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \cdots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}.$$

The system is called *homogeneous* if the *independent term* $\mathbf{b}$ is the null vector. A *solution* of the system above is a vector $\mathbf{x}^* = (x_1^*, \dots, x_n^*)^T$ satisfying the matrix equation, $A\mathbf{x}^* = \mathbf{b}$.

Definition 1.1. A linear system of equations is *consistent* if the system admits solutions. Otherwise it is called *inconsistent* or *overdetermined*. A consistent linear system with multiple solutions is called *underdetermined*.

Indeed, if a linear system has more than one solution, then it has infinitely many solutions.

Theorem 1.2. *If the linear system is underdetermined, then it admits infinitely many solutions.*

For $m \times 2$ systems there is a geometric interpretation. Each equation represents a straight line in the plane and the solutions of the system (if any) are the points of intersection of the m lines. In the case of two lines, $m = 2$, the system is consistent when the two lines intersect at a single point. The system is inconsistent if the two lines are parallel and not coincide. Finally, the system is underdetermined if the two lines coincide, that is, if the two equations are proportional.

Example 1.3. The system

$$\begin{cases} 2x + y = 5 \\ 4x + 2y = 7 \end{cases}$$

is inconsistent, because both lines are parallel (they have the same slope -2) but are not coincident, as the equations are not proportional.

The system

$$\begin{cases} x + y = 5 \\ 4y = 8 \end{cases}$$

is consistent with only one solution since they have different slopes -1 and 0 , respectively. Actually, the only solution is $(3, 2)$.

The system

$$\begin{cases} x + y = 4 \\ 2x + 2y = 8 \end{cases}$$

is underdetermined. Indeed, the equations represent the same line. One can describe the set of solutions as $\{(x, 4 - x) \mid x \in \mathbb{R}\}$.

2. ROUCHÉ–FROBENIUS THEOREM

Given a linear system $A\mathbf{x} = \mathbf{b}$, A will be called the *matrix of the system*, and the matrix $(A|\mathbf{b}) \in M_{m \times (n+1)}$ which contains A in the first n columns and $\mathbf{b}$ in column $n+1$ will be called the augmented matrix.

We will establish a criterium for studying the solutions of a general linear system in terms of the rank of the matrix of the system and the augmented matrix. Since every determinant in A also occurs in $(A|\mathbf{b})$, the rank of A cannot exceed that of $(A|\mathbf{b})$. Hence two possibilities exist: $\text{rank } A < \text{rank}(A|\mathbf{b})$ or $\text{rank } A = \text{rank}(A|\mathbf{b})$.

Theorem 2.1 (Rouché–Frobenius Theorem). *Consider the linear system $A\mathbf{x} = \mathbf{b}$. The system is consistent if and only if $\text{rank } A = \text{rank}(A|\mathbf{b})$. For a consistent system, we have*

- (1) *The system has only one solution if and only if the rank of A is equal to the number of unknown variables, $\text{rank } A = n$;*
- (2) *The system is underdetermined if and only if the rank of A is smaller than the number of unknown variables, $\text{rank } A = r < n$. Moreover, in this case the number of parameters needed to describe the solutions is $n - r$.*

Notice that a homogeneous system always has a solution, the trivial solution $\mathbf{0} = (0, \dots, 0)$. Indeed, $\text{rank } A = \text{rank}(A|\mathbf{0})$.

Example 2.2. Study the system

$$\begin{cases} x + y + 2z = 1 \\ 2x + y + 3z = 2 \\ 3x + 2y + 5z = 3 \end{cases}.$$

SOLUTION: Observe that $(A|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 2 & 1 & 3 & 2 \\ 3 & 2 & 5 & 3 \end{array} \right)$. We will find the rank of matrices A and $(A|\mathbf{b})$ simultaneously by means of elementary transformations.

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 2 & 1 & 3 & 2 \\ 3 & 2 & 5 & 3 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Thus, $\text{rank } A = \text{rank}(A|\mathbf{b}) = 2$. According to the previous theorem, the system is consistent, and the number of parameters in the solution set is $3 - \text{rank } A = 3 - 2 = 1$. We will show in the next section how to find the solution set.

2.1. The method of Gauss to solve linear systems. The method of Gauss to solve linear systems consists in finding the echelon form of the augmented matrix $(A|\mathbf{b})$. Then, the equivalent system is solved for this simpler equivalent system, from the last equation to the first one. Let us see this with an example.

Example 2.3. Find the solutions of the system

$$\begin{cases} x + y + 2z = 1 \\ 2x + y + 3z = 2 \\ 3x + 2y + 5z = 3. \end{cases}$$

SOLUTION: We already know that the system is consistent underdetermined and the echelon form of the augmented matrix is (see example above)

$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right).$$

Notice that the equivalent system is $\begin{cases} x + y + 2z = 1 \\ y + z = 0 \end{cases}$, which can be easily solved, starting by the last equation, $y = -z$ and substituting this into the first equation we get $x = 1 - y - 2z = 1 - (-z) - 2z = 1 - z$. Then, z plays the role of a parameter and the solution set is one-parametric, $\{(1 - z, -z, z) \mid z \in \mathbb{R}\}$, in accordance with our comments in the previous example.

2.2. Cramer's Rule. We study here a method based in determinants to give explicitly the solution for linear systems of order $n \times n$.

Theorem 2.4. *A linear system of order $n \times n$ is consistent with a unique solution if and only if $|A| \neq 0$.*

This is an easy consequence of the Rouché–Frobenius Theorem, since that when $|A| \neq 0$, $\text{rank } A = n$. Since it is always true that $\text{rank } A \leq \text{rank}(A|\mathbf{b}) \leq n$, necessarily $\text{rank } A = \text{rank}(A|\mathbf{b}) = n$ and the system is consistent with a unique solution.

Let a consistent system

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \vdots \\ a_{n1}x_1 + \cdots + a_{nn}x_n = b_n. \end{cases}$$

We know

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} \neq 0.$$

Let $(x_1^*, x_2^*, \dots, x_n^*)$ the unique solution of the system. Then, Cramer's rule gives the solution as

$$x_1^* = \frac{\begin{vmatrix} b_1 & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ b_n & a_{n2} & \cdots & a_{nn} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}}, \quad x_2^* = \frac{\begin{vmatrix} a_{11} & b_1 & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & b_n & \cdots & a_{nn} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}}, \dots, x_n^* = \frac{\begin{vmatrix} a_{11} & a_{12} & \cdots & b_1 \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & b_n \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}}.$$

Example 2.5. Find the solutions of the system

$$\begin{cases} 2x - 3y + 4z = 3 \\ 4x + y + z = 6 \\ 2x - y + z = 2. \end{cases}$$

SOLUTION: $|A| = \begin{vmatrix} 2 & -3 & 4 \\ 4 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} = -14 \neq 0$. Hence, by the above theorem, there is a unique

solution.

$$\begin{vmatrix} 3 & -3 & 4 \\ 6 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} = -14, \quad \begin{vmatrix} 2 & 3 & 4 \\ 4 & 6 & 1 \\ 2 & 2 & 1 \end{vmatrix} = -14, \quad \begin{vmatrix} 2 & -3 & 3 \\ 4 & 1 & 6 \\ 2 & -1 & 2 \end{vmatrix} = -14.$$

$x^* = 1$, $y^* = 1$, and $z^* = 1$.